

PAUL KARONJI WAITHAKA

Full-Stack Software Engineer · Systems Architect · Founder

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer and Systems Architect with 3+ years delivering production-grade systems spanning distributed backends, enterprise cloud platforms, offline-first desktop/mobile architectures, and cryptographic tooling. Founder of WIK Technologies, creator of **WIK Online Cloud POS** (NestJS 11 + KRA eTIMS fiscal compliance), **FleetTrack** (high-concurrency Go IoT telematics), **HotBill KE** (MikroTik network automation & M-Pesa billing), **MtaaKeys** (property marketplace monorepo), **DueSync** (Next.js SaaS + 13-tool native MCP AI server), and **Padi** (zero-knowledge PBKDF2 password manager). Deep expertise in Go, TypeScript, Python, and PHP, offline-first SQLite synchronization, emerging-market payment rails (M-Pesa, Paystack), and end-to-end production delivery. Final-year Software Development student at KCA University (Expected 2026).

TECHNICAL COMPETENCIES

- **Languages:** Go (Golang), TypeScript, JavaScript (ES6+), Python, PHP 8, Kotlin, SQL (PostgreSQL, MySQL, SQLite), HTML5, CSS3, Bash
- **Systems & Backend:** Go Goroutines & Channels, NestJS 11, Node.js, Express 5, Laravel 12, Next.js API Routes, WebSockets, Socket.IO, MikroTik RouterOS API (:8728), RESTful APIs
- **Frontend & Mobile:** React 19, Next.js 16/15, React Native 0.86, Expo 57, Electron, Vite, Tailwind CSS, Radix UI, TanStack Query, Progressive Web Apps (PWAs), IndexedDB
- **Databases & Storage:** PostgreSQL, MySQL 8 / TiDB Serverless, SQLite (better-sqlite3, Expo SQLite), Prisma ORM, Drizzle ORM, Eloquent ORM, Supabase, Cloudflare R2, AWS S3, Upstash Redis
- **Security & Cryptography:** Zero-Knowledge client-side encryption (Web Crypto PBKDF2/SHA-256 100k rounds, AES-GCM-256), hardware-bound CPU/MAC fingerprinting, timing-safe HMAC-SHA256, RLS, Helmet, rate limiting, KRA eTIMS AES-256 keys, CVE patching
- **DevOps & Infrastructure:** Docker, Docker Compose, Nginx edge reverse proxy, Vercel, Oracle Cloud Infrastructure (ARM Ampere A1), GitHub Actions CI/CD, Vercel Cron, Netlify Functions
- **Payments & Fiscal Compliance:** KRA eTIMS (OSCU/VSCU automated fiscalization), M-Pesa Daraja (STK Push & C2B Webhook queues), Paystack (SaaS subscriptions & split payments), Model Context Protocol (MCP), Google Calendar API v3 (OAuth 2.0), Africa's Talking, Cloudinary, Resend

PROFESSIONAL EXPERIENCE

Founder & Systems Architect | WIK Technologies *Nairobi, Kenya · Oct 2025 – Present*

- **WIK Online Cloud POS:** Architected multi-tenant cloud retail operations platform on NestJS 11, React 19, and PostgreSQL; built automated KRA eTIMS fiscalization worker (OSCU/VSCU) with AES-256 encrypted tax credentials, real-time M-Pesa STK push & C2B Paybill cashier reconciliation queue (C2BQueueModal), and Paystack recurring SaaS billing with lifecycle walls (ActiveSubscriptionGuard, SuspendedWall).
- **Brixton Makunga Hospital Management System:** Delivered 200+ endpoint clinical hospital ERP on Laravel 12, React 19, and MySQL 8 with automated ledger billing via Observer pattern, real-time triage queue management, and pharmacy FEFO batch expiry registers.
- **FleetTrack Enterprise Telematics:** Engineered high-concurrency Go backend for batch IoT telematics ingestion and real-time WebSocket fan-out, paired with an offline-first React Native (Expo 57) driver mobile app buffering telemetry in local SQLite with QR manifest claiming and stoppage detection.
- **HotBill KE Telecommunications:** Built automated telecommunications billing daemon integrating MikroTik RouterOS socket API with M-Pesa Daraja STK push and a Next.js 14 NOC console for ISP/hotspot bandwidth automation in Docker Compose.
- **Padi Password Manager:** Developed zero-knowledge cross-browser extension (Manifest V3) and Web Vault using client-side Web Crypto PBKDF2 (100,000 iterations) and AES-GCM-256 with Supabase synchronization.

- **WIK-POS Desktop Suite & Licensing Authority:** Shipped offline-first Electron + SQLite retail suite featuring hardware-bound licensing (CPU+MAC fingerprinting), 30-day offline grace periods, FEFO pharmacy expiry batching (v1.3.33), and hotel multi-terminal LAN F&B ordering (v1.3.26).
- **DueSync Smart Task Management & MCP:** Shipped task-management SaaS on Next.js 15 featuring two-way Google Calendar OAuth2 sync, Upstash Redis rate limiting, and a native 13-tool MCP server allowing AI agents to orchestrate schedule planning.

Technical Lead (Industrial Attachment) | JHUB Africa *Nairobi, Kenya · Sep 2025 – Dec 2025*

- **JHUB Africa Innovation Tracker:** Directed engineering of the JKUAT innovation management platform (PHP 8.1, MySQL, Docker, JavaScript) across 4 user roles, 6 project lifecycle stages, and automated mentor assignment engines; delivered in production for active nationwide use.
- **Quality & DevOps Standards:** Established PSR-12 coding standards, conducted code reviews, and containerized the platform with Docker Compose, ensuring high reliability for innovators and investors.

Freelance Software Engineer & Systems Consultant | Independent Contractor *Remote & Nairobi · 2023 – Present*

- **MtaaKeys Property Marketplace:** Shipped full-stack property marketplace monorepo (Next.js 16, Express 5, PostgreSQL, Prisma, Socket.IO) with dual Paystack subscription tiers (agents & land agents), logarithmic listing curation algorithms, and CI/CD security invariant tests.
- **K-Track Commission SaaS:** Built multi-tutor agency commission platform (React 19, Express 5, TiDB Serverless, Cloudflare R2, Socket.IO) with timing-safe HMAC guest checkout URLs and idempotent boot-time database patching.
- **BikeLab Parts Manager:** Built desktop parts catalog and workshop inventory manager with Electron, Vite, React, and local SQLite, providing high-performance hardware acceleration for warehouse cataloging.
- **Biashara Hub Headless Commerce:** Engineered production headless e-commerce for Kenyan retail using Medusa.js v2 and Next.js App Router with native M-Pesa payment rails.

EDUCATION & CREDENTIALS

Bachelor of Science in Software Development | KCA University *Nairobi, Kenya · Sep 2021 – Dec 2026 Expected*

- **Relevant Coursework:** Advanced Web Development, Data Structures & Algorithms, Database Management, Network Security, Software Engineering, System Integration, Mobile Application Development, OOP, Agile Project Management.
- **Certifications:** Cybersecurity Fundamentals (IBM SkillsBuild, 2025) · Data Protection Basics (Atingi eLearning Platform, Sep 2025) · Zoho Creator Developer Certification (Zoho, Nov/Dec 2025).